

QUESTION 1

Code:

```
# Data
time <- c(18,21,22,24,25,25,27,28,29,30,
          31,32,33,35,36,38,41,43,46,50)
# 1. Histogram
hist(
  time,
  breaks = 6,
  main = "Histogram of Processing Times",
  xlab = "Processing Time (minutes)",
  ylab = "Frequency",
  col = "lightblue",
  border = "white"
)
# 2. Density Plot
density_time <- density(time)

plot(
  density_time,
  main = "Density Plot of Processing Times",
  xlab = "Processing Time (minutes)",
  ylab = "Density"
)
polygon(
  density_time,
```

```

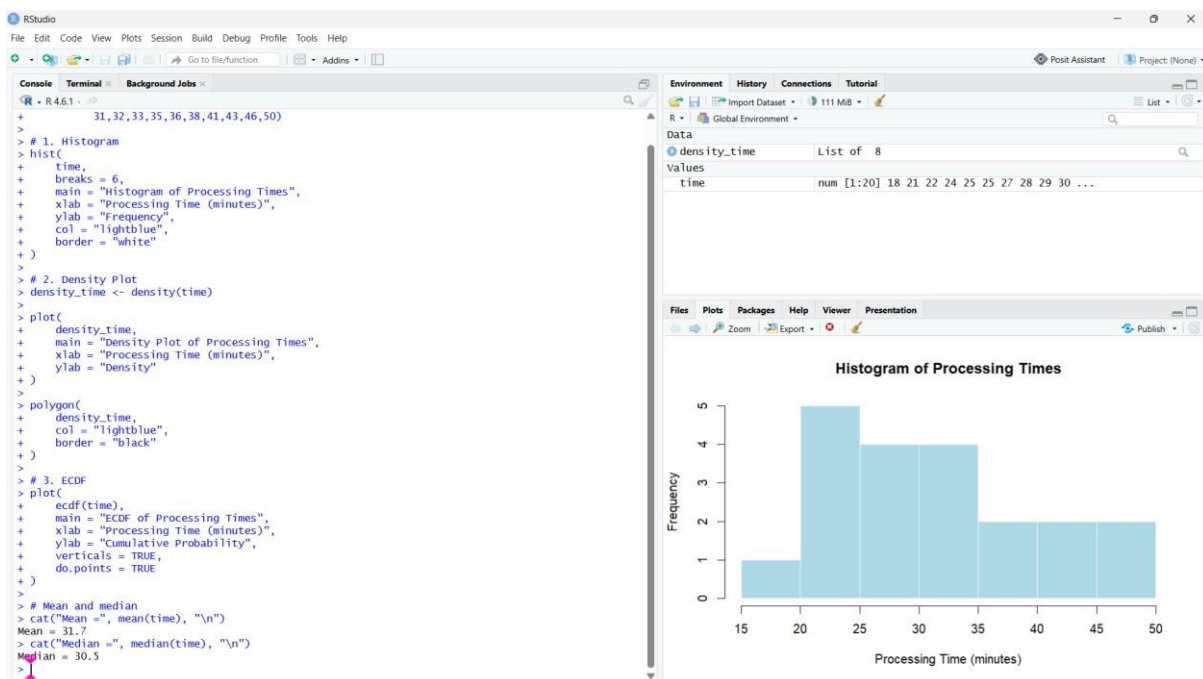
col = "lightblue",
border = "black"
)
# 3. ECDF

plot(
  ecdf(time),
  main = "ECDF of Processing Times",
  xlab = "Processing Time (minutes)",
  ylab = "Cumulative Probability",
  verticals = TRUE,
  do.points = TRUE
)
# Mean and median
cat("Mean =", mean(time), "\n")
cat("Median =", median(time), "\n")

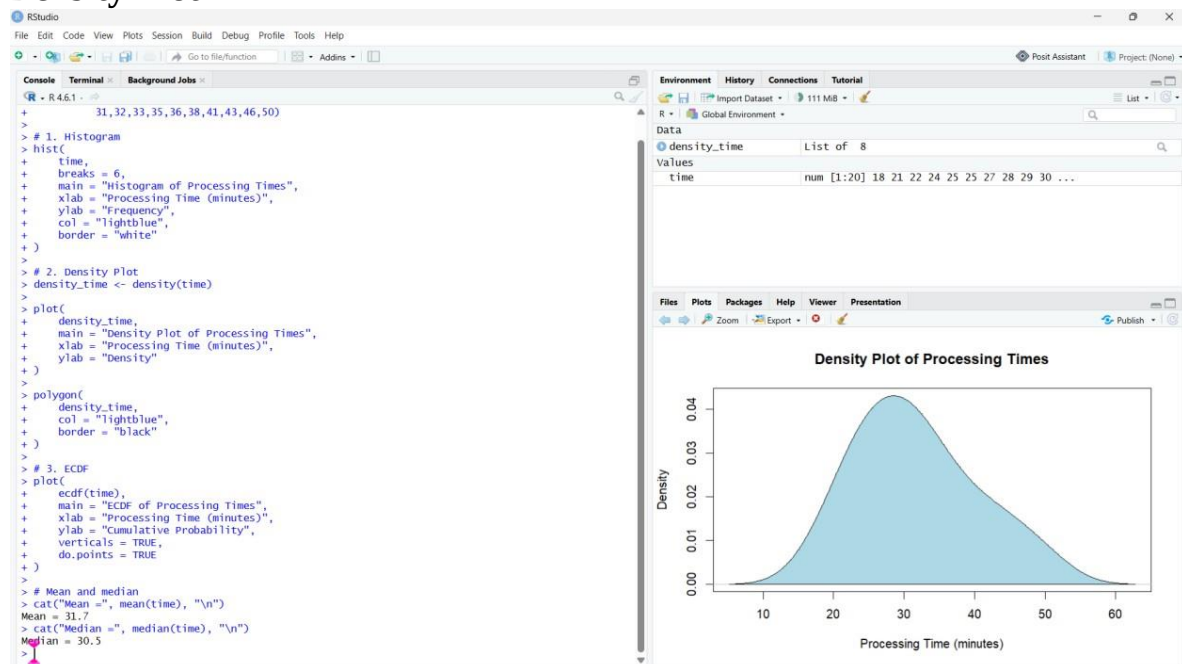
```

Output:

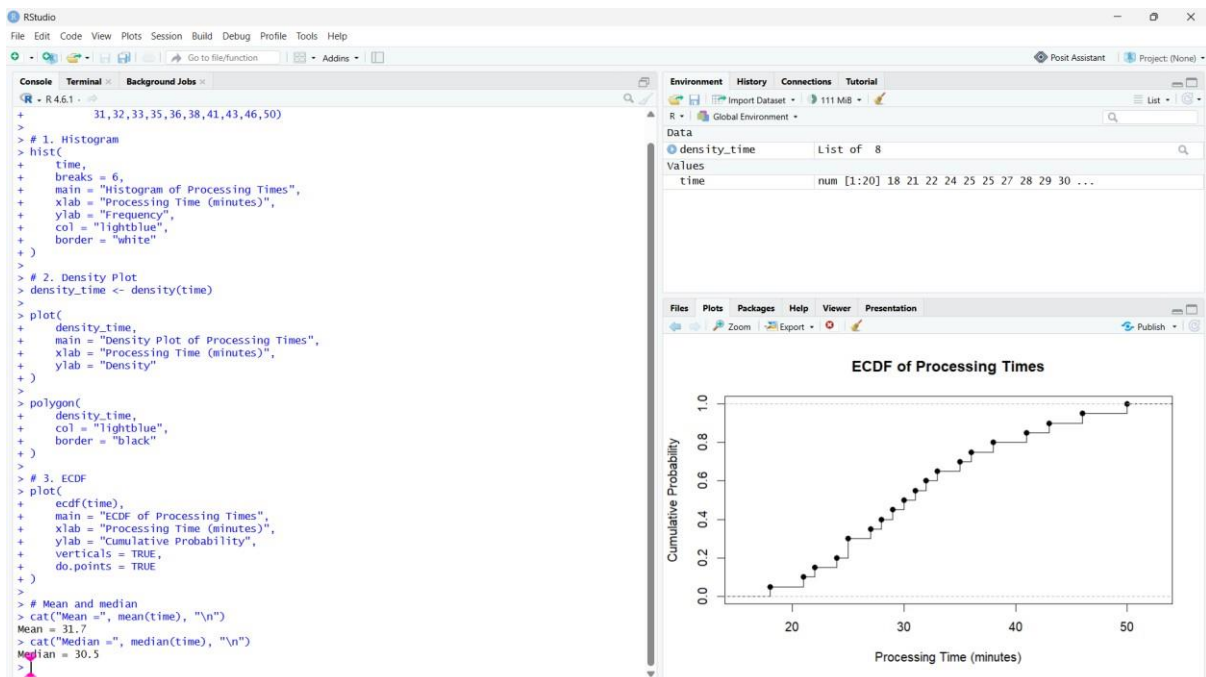
Histogram



Density Plot



ECDF



QUESTION 2

Code:

```
# Data
response <- c(42,45,47,49,50,51,52,53,54,55,
              56,57,58,60,61,63,66,70,76,85)

# 1. Normal Q-Q Plot
qqnorm(
  response,
  main = "Normal Q-Q Plot of Response Times",
  xlab = "Theoretical Quantiles",
  ylab = "Observed Response Times"
)
qqline(
  response,
```

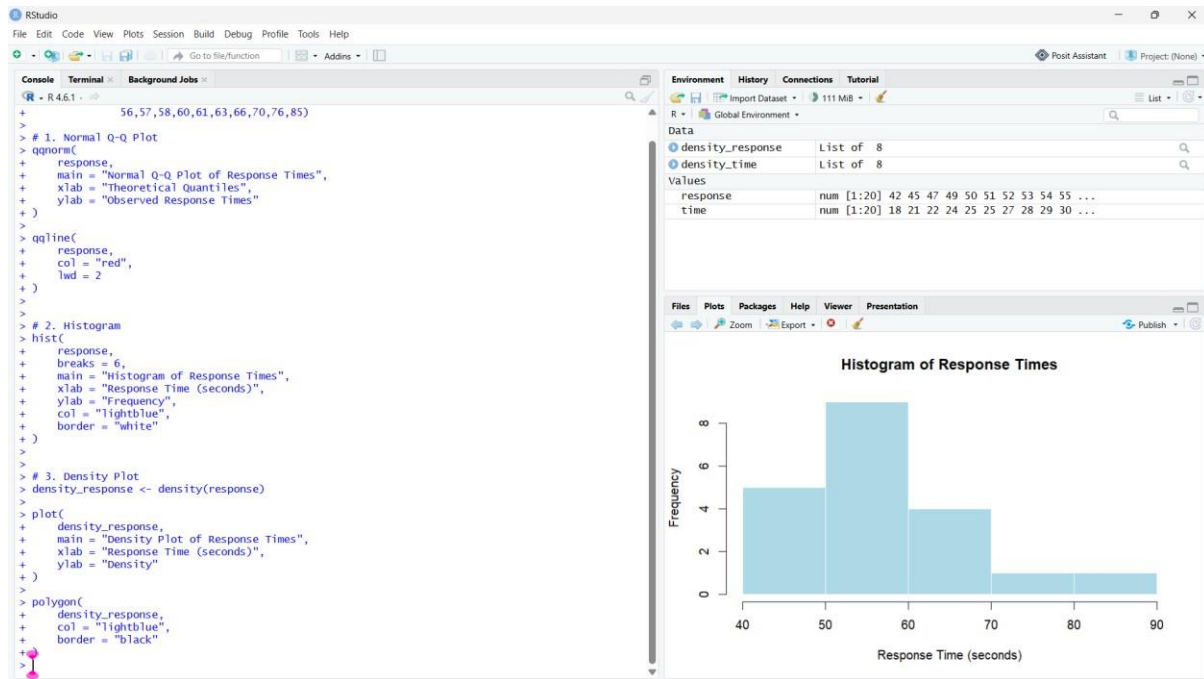
```
col = "red",  
lwd = 2  
)  
# 2. Histogram  
hist(  
  response,  
  breaks = 6,  
  main = "Histogram of Response Times",  
  xlab = "Response Time (seconds)",  
  ylab = "Frequency",  
  col = "lightblue",  
  border = "white"  
)  
# 3. Density Plot  
density_response <- density(response)  
plot(  
  density_response,  
  main = "Density Plot of Response Times",  
  xlab = "Response Time (seconds)",  
  ylab = "Density"  
)  
polygon(  
  density_response,  
  col = "lightblue",
```

border = "black"

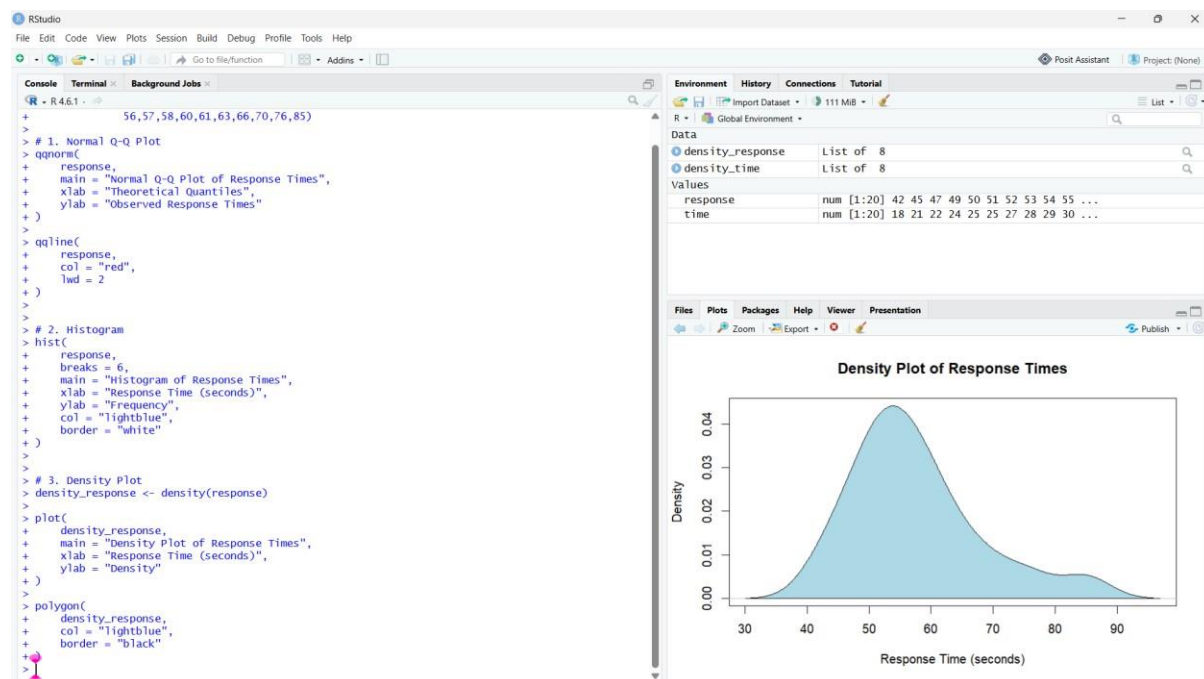
)

Output:

Histogram



Density plot:



Interpretation

- Most observations lie reasonably close to the **Q-Q reference line**, suggesting an approximately normal distribution in the central region.
- The **upper tail deviates noticeably above the reference line**, especially around **76 and 85 seconds**.
- This indicates **right-skewness and possible extreme observations** in the upper tail.
- The histogram/density plot also shows a longer **right tail**, supporting the Q-Q plot's conclusion.

QUESTION 3

Code:

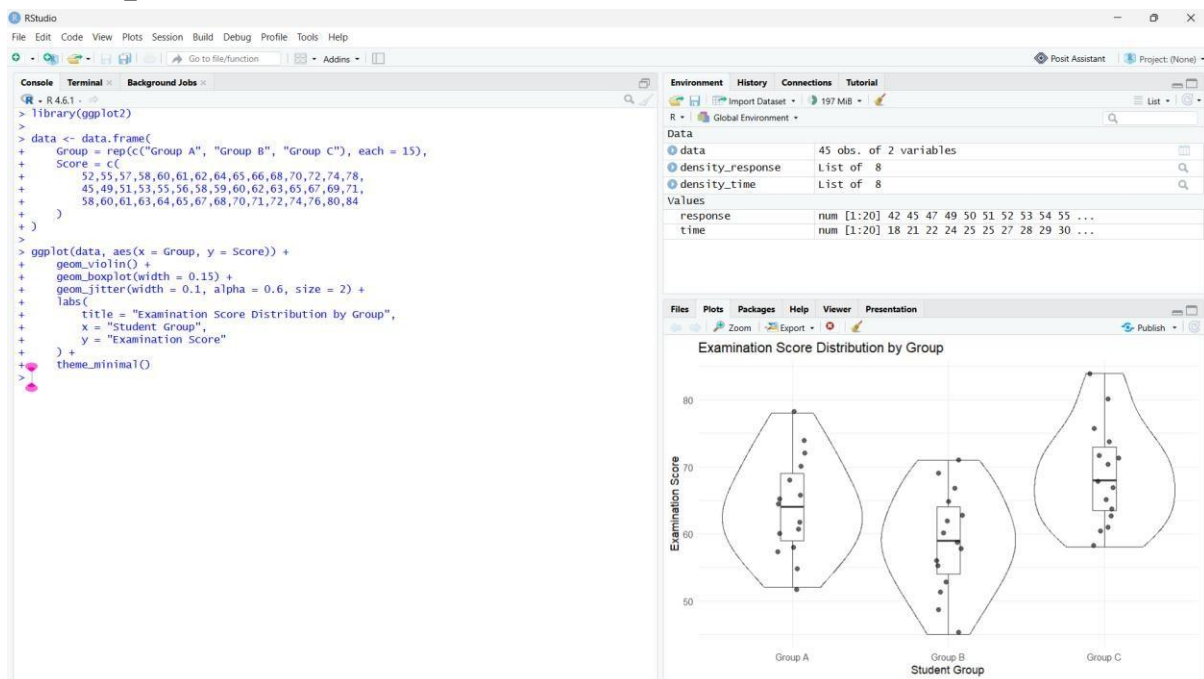
```
library(ggplot2)
data <- data.frame(
  Group = rep(c("Group A", "Group B", "Group C"), each = 15),
  Score = c(
    52,55,57,58,60,61,62,64,65,66,68,70,72,74,78,
    45,49,51,53,55,56,58,59,60,62,63,65,67,69,71,
    58,60,61,63,64,65,67,68,70,71,72,74,76,80,84
  )
)
ggplot(data, aes(x = Group, y = Score)) +
  geom_violin() +
```

```

geom_boxplot(width = 0.15) +
geom_jitter(width = 0.1, alpha = 0.6, size = 2) +
labs(
  title = "Examination Score Distribution by Group",
  x = "Student Group",
  y = "Examination Score"
) +
theme_minimal()

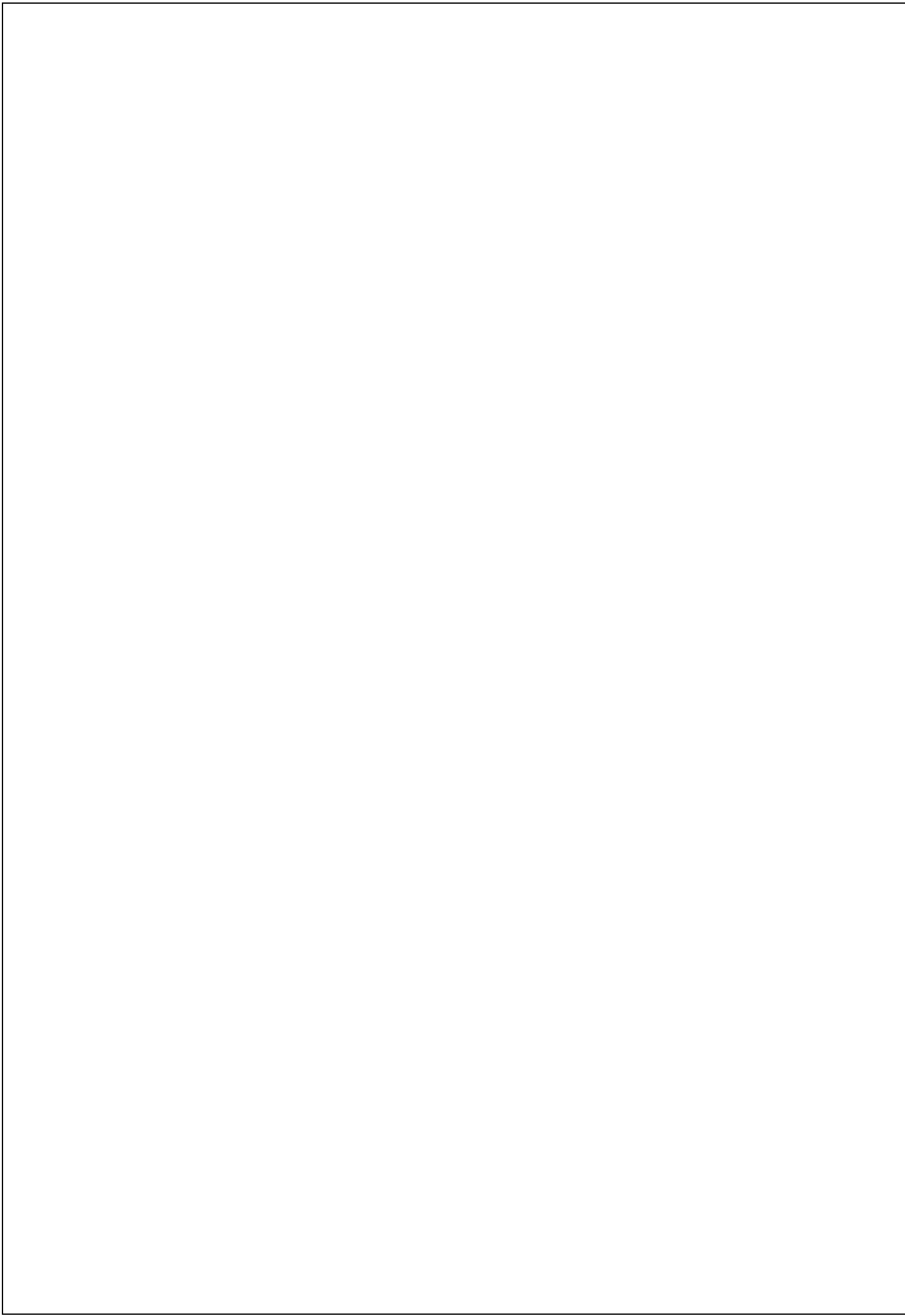
```

Output:



Interpretation

- **Group A** has a center around the mid-60s with moderate spread.
- **Group B** has the **lowest center**, with scores ranging from 45 to 71.
- **Group C** has the **highest center** and extends up to 84, showing relatively high performance.
- There are **no obvious extreme outliers** in the three groups.



QUESTION 4

Code:

```
library(ggplot2)
data <- data.frame(

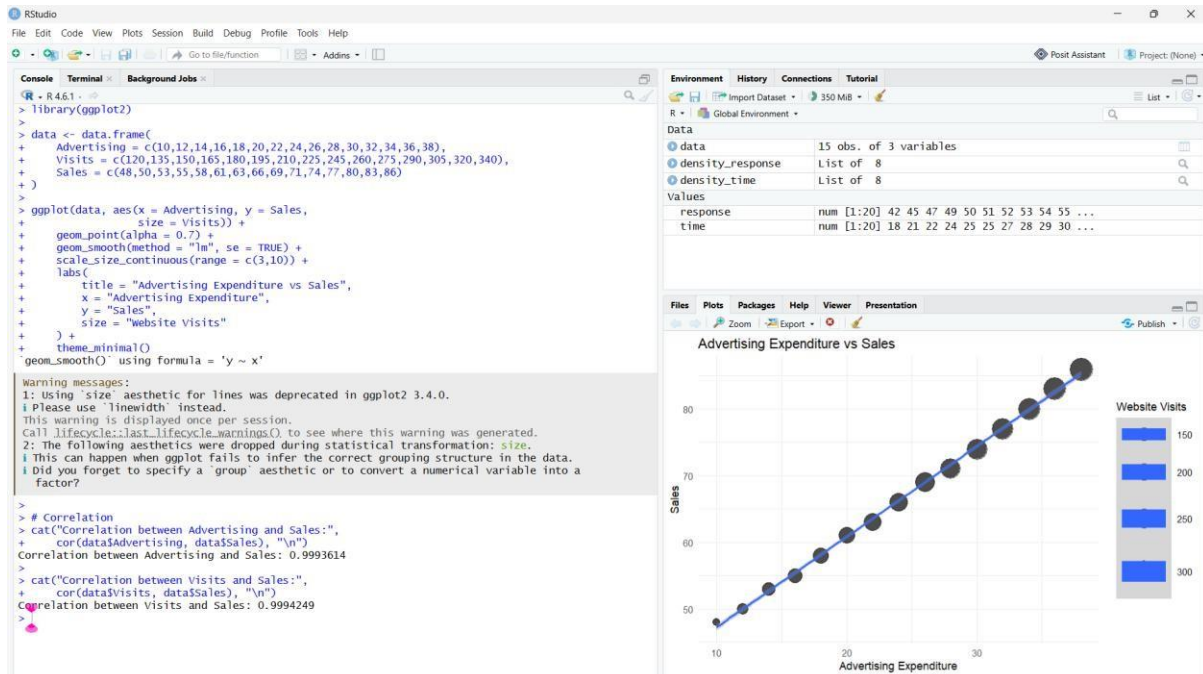
  Advertising = c(10,12,14,16,18,20,22,24,26,28,30,32,34,36,38),

  Visits = c(120,135,150,165,180,195,210,225,245,260,275,290,305,320,340),

  Sales = c(48,50,53,55,58,61,63,66,69,71,74,77,80,83,86)

)
ggplot(data, aes(x = Advertising, y = Sales,
                 size = Visits)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  scale_size_continuous(range = c(3,10)) +
  labs(
    title = "Advertising Expenditure vs Sales",
    x = "Advertising Expenditure",
    y = "Sales",
    size = "Website Visits"
  ) +
  theme_minimal()
# Correlation
cat("Correlation between Advertising and Sales:",
    cor(data$Advertising, data$Sales), "\n")
cat("Correlation between Visits and Sales:",
    cor(data$Visits, data$Sales), "\n")
```

Output:



Interpretation

- Advertising expenditure and sales show a **strong positive relationship**.
- Website visits are also positively related to sales.
- Using **bubble size** for website visits adds useful information without requiring another plot.

QUESTION 5

Code:

```
library(ggplot2)
data <- data.frame(
```

```
  Department = c("CSE","ECE","EEE","MECH","CIVIL","AIDS","IT"),
```

```
  Pass_Percentage = c(91,93,92,89,90,95,94)
```

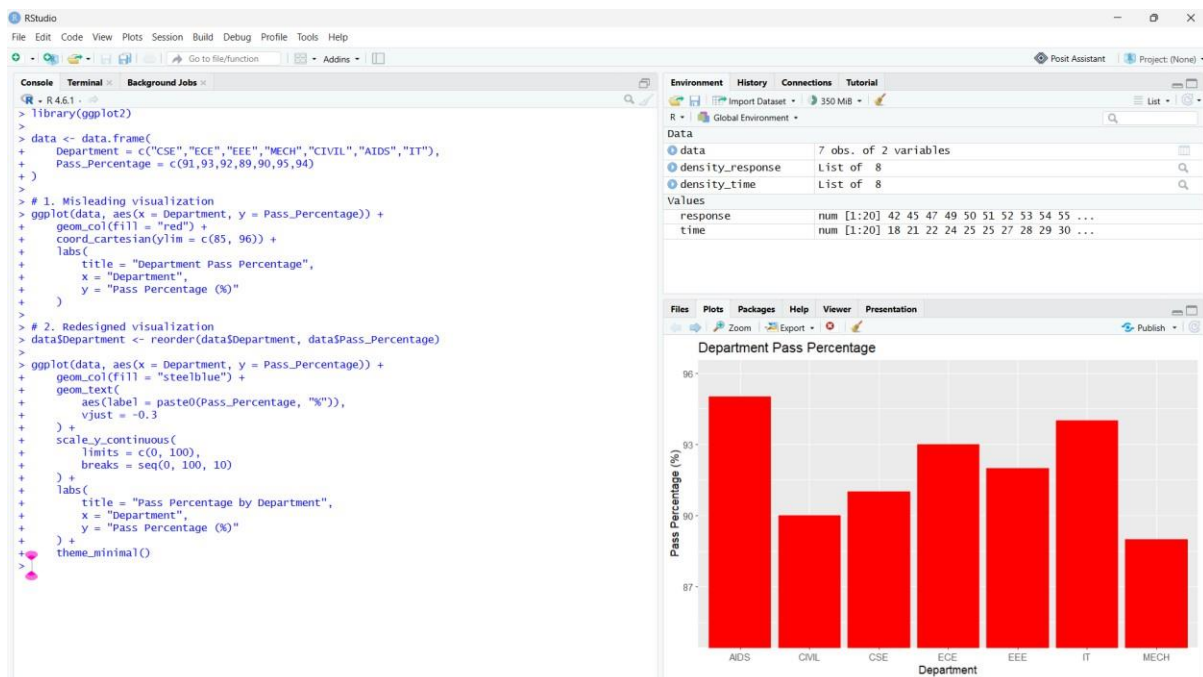
```
)
# 1. Misleading visualization
ggplot(data, aes(x = Department, y = Pass_Percentage)) +
  geom_col(fill = "red") +
  coord_cartesian(ylim = c(85, 96)) +
  labs(
    title = "Department Pass Percentage",
    x = "Department",
    y = "Pass Percentage (%)"
  )
```

```
# 2. Redesigned visualization
data$Department <- reorder(data$Department, data$Pass_Percentage)
ggplot(data, aes(x = Department, y = Pass_Percentage)) +
  geom_col(fill = "steelblue") +
  geom_text(
    aes(label = paste0(Pass_Percentage, "%")),
    vjust = -0.3
  ) +
  scale_y_continuous(
    limits = c(0, 100),
    breaks = seq(0, 100, 10)
  ) +
  labs(
    title = "Pass Percentage by Department",
    x = "Department",
    y = "Pass Percentage (%)"
  ) +
  theme_minimal()
```

Interpretation

- The **first graph** uses a truncated y-axis starting at 85%, which makes small differences such as 89% vs 95% appear much larger.
- The second graph starts at **0%**, giving a more honest visual comparison.
- Departments are **ordered by pass percentage**, making ranking easier to understand.

Output: Pass Percentage



Passenger by departed

